A Survey of Question Answering Using Semantic Roles

SRL-Based QA Systems, PropBank, and Beyond

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Abstract

This survey reviews semantic role labeling (SRL) as a representational substrate for question answering (QA), with PropBank serving as the primary anchor between linguistic theory and practical engineering. The core argument is that SRL-based QA remains relevant because it makes predicate-argument structure explicit: instead of only predicting an answer span, the system can explain which event slot a question targets. We therefore synthesize the feature-engineering era, the neural sequence-labeling era, and the current transformer and LLM era through the lens of answerability, interpretability, and corpus design.

The survey is grounded by a concrete local PropBank run. In the current environment, NLTK exposes 112,917 PropBank instances overall, while parse-tree access is available for 9,073 instances in the bundled Treebank subset, yielding 23,007 automatically generated QA examples. This observation matters because it highlights a persistent theme in the literature: practical SRL-QA quality depends not only on model design, but also on how annotation, parse access, and argument-span recoverability are packaged. Across the literature, the most robust findings concern the usefulness of argument-aware abstraction, the naturalness of wh-question mappings, and the continuing need for calibrated, frame-aware reasoning in complex settings.

Introduction

Question answering systems are usually judged by whether they return the right string, yet many high-stakes applications also require a principled explanation of why that string counts as the answer. SRL-based QA addresses this need by decomposing a sentence into predicates and arguments before answering. In effect, the question becomes a request for a semantic slot: who performed the event, what entity was affected, when it occurred, where it happened, how it unfolded, or why it took place.

This perspective is attractive because it aligns directly with information extraction and event understanding. PropBank, in particular, packages semantic roles in a way that is coarse enough to scale and precise enough to map onto natural-language questions. That is why the survey centers PropBank while also contrasting it with FrameNet, VerbNet, QA-SRL, OntoNotes, and CoNLL resources.

The survey proceeds from background and corpus design to historical system development, then concentrates on direct SRL-QA work and closes with research gaps. Throughout, recent 2022-2025 literature is used to update the story beyond the classical SRL canon and to motivate the innovation proposals that accompany this project.

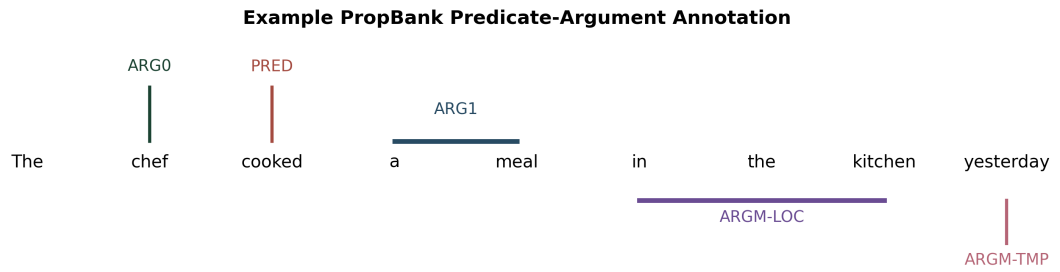

Figure 1. PropBank-style predicate-argument annotation as a basis for QA generation.

Background: Semantic Role Labeling

SRL formalizes event structure by assigning roles to the participants and modifiers associated with a predicate. It is often described as shallow semantics because it sits between syntax and full logical form: richer than phrase structure alone, but less brittle than complete symbolic semantic parsing. This middle position is precisely what makes SRL useful for QA. It offers a constrained answer space without requiring a question to be translated into a fully formal program.

PropBank's design is especially pragmatic. Numbered core arguments are defined locally at the roleset level, so ARG0 is often agent-like and ARG1 often patient-like, but the precise interpretation is anchored in the predicate sense. Adjunct roles such as ARGM-TMP, ARGM-LOC, ARGM-MNR, ARGM-CAU, and ARGM-PRP naturally support when, where, how, why, and purpose questions. Compared with FrameNet's more semantically expressive frame elements or VerbNet's class-based generalizations, PropBank is less elegant but highly portable.

This portability is one reason SRL became a durable engineering abstraction. It allows a system to normalize across passives, alternations, and surface realizations while keeping role recovery auditable. That same property explains why QA-based semantics repeatedly returns to question templates as an annotation and inference interface.

Role family	Typical interpretation	Representative QA mapping
ARG0	Proto-agent, experiencer, speaker, causer	Who did the action?
ARG1	Patient, theme, proposition, affected entity	What was affected or asserted?
ARG2-ARG5	Predicate-specific secondary roles	To whom, from where, with what?
ARGM-TMP	Temporal modifier	When did it happen?
ARGM-LOC / DIR / GOL	Spatial modifier	Where did it happen?
ARGM-MNR / ADV / EXT	Manner or extent	How did it happen?
ARGM-CAU / PRP / PNC	Cause or purpose	Why did it happen?

Table 1. PropBank role families and their QA interpretations.

PropBank Corpus - Structure and Statistics

PropBank augments the Penn Treebank with predicate-argument annotations. NLTK exposes this through `propbank.instances()` for annotated predicates and `propbank.roleset()` for frame files, making the corpus attractive for reproducible coursework and compact research prototypes. The trade-off, documented explicitly in the NLTK PropBank how-to, is that the standard downloadable Treebank sample contains parse trees for only the initial portion of the full corpus.

In the present run, this design yields 112,917 PropBank instances overall but only 9,073 Treebank-backed instances for full span reconstruction, which still support 23,007 QA examples. That is a useful reminder that corpus access constraints shape the effective data distribution seen by a model. They influence not only sample size, but also which predicate types, syntactic constructions, and answer spans are available at training time.

Roleset frame files remain indispensable because they define how numbered arguments should be interpreted for each predicate sense. For QA generation, those glosses help avoid overly generic templates and support better alignment between target role and wh-question type.

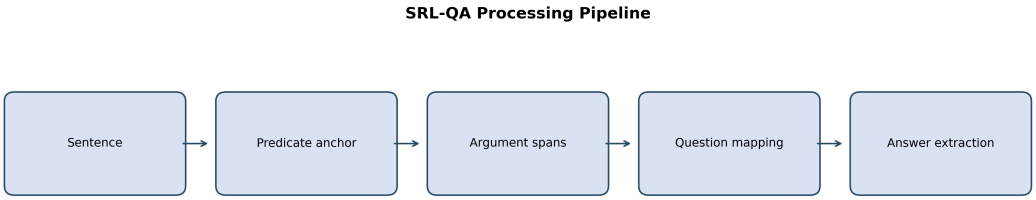

Figure 2. End-to-end SRL-QA pipeline from sentence structure to answer extraction.

SRL Systems - Historical and Modern Overview

The feature-engineering era established the problem. Gildea and Jurafsky (2002) showed that semantic roles could be learned statistically from parse-derived features, while the CoNLL shared tasks standardized evaluation and pushed the field toward better voice, path, phrase-type, and subcategorization features. These systems were heavily syntax-dependent, but they made the latent structure of the task explicit.

The neural era replaced much manual feature engineering with contextual encoders. Zhou and Xu (2015), He et al. (2017), and syntax-agnostic neural models demonstrated that recurrent encoders could recover much of the relevant context without hand-crafted templates. Span-based and self-attentive models then generalized the search space beyond fixed local decisions.

The transformer and LLM era changed the discussion again. Pretrained encoders improved SRL as a supervised task, while more recent work explored LLMs as adjudicators, prompt-driven SRL solvers, or frame-aware reasoners. Notably, current research often returns to explicit semantics when generic generation proves too unconstrained for reliable role analysis.

From an engineering standpoint, each historical phase changed what counted as the difficult part of SRL. In the feature era, the main challenge was designing informative abstractions over syntax. In the neural era, the challenge shifted to contextual representation and decoding. In the transformer and LLM era, the challenge is increasingly about grounding, calibration, and preserving semantic faithfulness in the presence of highly fluent but potentially underspecified models.

This historical comparison is especially relevant to QA because answer extraction inherits the weaknesses of the underlying SRL layer. A brittle parser yields brittle slots; an opaque encoder yields hard-to-explain answers. The best SRL-QA systems therefore balance representational power with enough structure to keep errors interpretable.

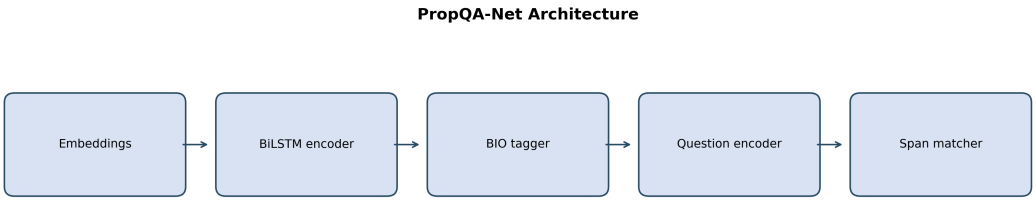

Figure 3. A compact view of the PropQA-Net architecture used in the empirical companion study.

Question Answering Systems - Overview

Reading-comprehension QA, knowledge-base QA, and open-domain QA each optimize a different slice of the answering problem. SRL-based QA shares the extractive bias of reading comprehension but constrains the answer search space using event structure. This makes it especially attractive for event-centric information extraction, auditable decision support, and settings where wh-question types map naturally onto semantic slots.

The limitation is equally clear: SRL-QA inherits the blind spots of SRL. Multi-predicate reasoning, long-distance dependencies, discourse phenomena, and genre transfer remain challenging unless additional layers of reasoning are added.

SRL-Based QA - Related Work

Direct SRL-QA research spans several sub-traditions. Shen and Lapata (2007) used semantic roles to improve answer selection; He et al. (2015) reformulated SRL annotation itself as question answering; Michael et al. (2018) and FitzGerald et al. (2018) scaled question-answer-based meaning representations and parsing. Later work extended the paradigm to nominalizations (QANom), discourse relations (QADiscourse), question generation (Asking It All), cross-text alignment (QA-Align), text-to-text semantic parsing (QASem), and adjectival semantics (QA-Adj).

Recent work shows that the paradigm is still evolving. Bonn et al. (2024) tested LLMs as PropBank adjudicators, Chen et al. (2025) argued that LLMs can perform well on SRL when guided structurally, and Rai et al. (2025) injected frame semantics into LLMs via prompt-based fine-tuning. These efforts suggest that explicit semantics is becoming more, not less, relevant in the LLM era.

A striking pattern across these papers is that question formulation itself acts as an inductive bias. Systems improve when the question reveals the intended semantic contrast clearly, whether by specifying an event participant, a modifier type, or a cross-text alignment objective. This is one reason the literature often oscillates between building better parsers and building better questions; the two are tightly coupled.

Another pattern is scope expansion. Early work treated QA primarily as an interface to verbal arguments. Recent work includes nominalizations, adjectives, discourse relations, alignment, and frame-semantic probing. The research space is therefore broadening from 'use SRL to answer questions' to 'use question-answer structure as a universal surface form for semantics.'

He et al. (2015) should also be understood as a methodology paper about annotation economics. By allowing annotators to ask and answer natural-language questions rather than memorize a formal label inventory, QA-SRL changed who could contribute high-quality semantic supervision. That design decision continues to influence later work on scalable semantic annotation.

FitzGerald et al. (2018) and Roit et al. (2020) jointly illustrate the transition from proof of concept to infrastructure. One showed that QA-SRL parsing can scale with neural models; the other showed that the supervision pipeline can scale without catastrophic quality loss. Together they established QA-based semantics as an engineering paradigm rather than only a crowdsourcing novelty.

QANom, QA-Adj, and QADiscourse are significant because they challenge the default assumption that predicate semantics is mostly verbal. Once question-answer form becomes the interface, nominal, adjectival, and discourse semantics become much easier to fold into the same representational family. This broadens the reach of SRL-QA beyond classic event extraction.

The most recent literature adds another twist: LLMs are increasingly used to support, rather than replace, explicit semantics. PropBank adjudication, frame-aware prompting, and structured QA probing all suggest that symbolic resources become more valuable when they can constrain or audit flexible language models.

Paper	Year	Method	Dataset	SRL Tool	QA Type	F1	EM	Key Innovation
Shen & Lapata	2007	Role-aware QA reranking	TREC-style QA	Classical SRL	Factoid	N/A	N/A	Early use of semantic roles for answer se
He et al.	2015	QA-SRL annotation schema	QA-SRL	Human QA roles	Annotation	N/A	N/A	Natural-language questions as role labels
Michael et al.	2018	QAMR crowdsourcing	QAMR	QA meaning reps	Semantic parsing	N/A	N/A	Open QA form for predicate-argument m
FitzGerald et al.	2018	Neural QA-SRL parser	QA-SRL Bank 2.0	Neural span model	Span QA	Reported in paper	N/A	Large-scale automatic QA-SRL parsing
Roit et al.	2020	Controlled crowdsourcing	QA-SRL	QA-SRL	Annotation	N/A	N/A	Improved label quality and scalability
Klein et al.	2020	Nominal QA-SRL	QANom	QA-SRL extension	Nominal QA	Reported in paper	N/A	Extends QA-SRL beyond verbs
Pyatkin et al.	2020	QA discourse relations	QADiscourse	QA semantics	Discourse QA	Reported in paper	N/A	QA representation for discourse links
Pyatkin et al.	2021	Question generation	QA-SRL / QASRL	QA-SRL	Question generation	Reported in paper	N/A	Generates contextualized questions for ar
Weiss et al.	2021	QA proposition alignment	QAMR Align	QA propositions	Cross-text QA	Reported in paper	N/A	Cross-document alignment via QA pairs
Klein et al.	2022	Text-to-text QA semantics	QASem	Seq2seq QA semantics	Semantic parsing	Reported in paper	N/A	Unified text-to-text formulation
Pesahov et al.	2023	Adjectival QA semantics	QA-Adj	QA-based semantics	Modifier QA	Reported in paper	N/A	Adds adjectival semantics
Bonn et al.	2024	LLM-assisted PropBank adjudication	PropBank	Prompted LLMs	Annotation support	Reported in paper	N/A	Tests LLMs for expert adjudication
Chen et al.	2025	LLM-based SRL	CoNLL / SRL corpora	LLM prompting	Closed-domain QA	Reported in paper	N/A	Shows LLMs can recover structured SRL
Rai et al.	2025	Frame-semantic LoRA	FrameNet	Frame-aware LLM	QA probing	Reported in paper	N/A	Injects frame knowledge to improve zero

Table 2. Representative work at the SRL-QA intersection.

Detailed Literature Capsules

QA-SRL's long-term significance lies in its reframing of semantics as a question space. That move changed both annotation workflows and the mental model of what a semantic parser can expose to end users: instead of opaque labels alone, the annotation itself becomes directly queryable.

QAMR broadened the perspective from role inventories to proposition-level meaning. This matters because many practical QA tasks are not only about slots inside a single predicate, but about broader meaning units that still benefit from QA-style decomposition.

Large-scale QA-SRL parsing demonstrated that the question-based representation is compatible with neural automation, which is what made later systems feasible at scale. Without that step, QA-based semantics might have remained primarily a crowdsourcing framework.

QANom is particularly important for deployment because many domains distribute key facts through nominalizations rather than verbal clauses. Contracts, scientific writing, and newswire all contain event information in noun phrases, and a purely verbal SRL-QA system would miss much of that content.

QADiscourse and QA-Align show that the same representation can travel upward from sentence-internal roles to discourse and cross-document structure. This suggests that QA-style semantics can become a bridge representation across multiple granularities of language understanding.

QASem and QA-Adj indicate a more general research direction: once semantic analysis is expressed as question-answer text, it can inherit the modeling advantages of text-to-text architectures without surrendering interpretability. That trend makes QA-based semantics especially compatible with current generation-oriented NLP workflows.

LLM-era work reintroduces the question of what must remain explicit. The recent adjudication and frame-injection papers imply that large models are most useful when a semantic scaffold already exists. In other words, the future may belong less to replacing SRL and more to instrumenting LLMs with SRL-like structure.

Across all these papers, the research gap is not simply accuracy. It is the absence of a single evaluation framework that rewards local role fidelity, cross-sentence reasoning, calibrated abstention, and explanation simultaneously. That is the gap the current project's innovation document attempts to fill.

Datasets Used in SRL-QA Research

PropBank is the most direct fit for classical SRL-QA because its labels are already aligned to predicate senses and answerable spans. QA-SRL and related resources instead use questions as the semantic representation itself, making them especially suitable for annotation and interface research. OntoNotes broadens genre coverage, CoNLL-2009 supports dependency-based and multilingual comparison, and FrameNet provides semantically richer but less uniform frame elements.

These datasets are not interchangeable. Choosing among them determines what kind of answer structure a model can learn, how easily questions can be generated, and whether evaluation emphasizes local argument recovery or broader semantic coverage. For this project, PropBank is valuable precisely because it sits at the intersection of linguistic supervision and practical extractive QA.

This comparison also clarifies why PropBank remains a sensible baseline even when richer resources exist. A benchmark built on named frame elements or cross-text propositions can be more semantically expressive, but it is harder to align with a simple, reproducible QA system. PropBank offers a strong

compromise between semantic clarity and implementation tractability.

Dataset	Primary annotation	Typical use	Availability	Relevance to this project
PropBank	Predicate-argument roles (ARG0-ARG5, ARGM)	SRL, ARGM grounding	Public frames; PTB-linked training source	Primary training source; local NLTK sample 112917 t
QA-SRL	Natural-language role questions	Role induction, parsing, QG	Public	Closest direct SRL-QA benchmark
OntoNotes 5.0	Syntax, entities, SRL, coreference	Robust SRL and IE	Licensed	Common neural SRL benchmark
CoNLL-2009	Dependency SRL	Multilingual SRL	Shared-task data	Useful for dependency-oriented comparison
FrameNet	Frames and frame elements	Frame semantics, grounding	Public	Useful for innovation beyond PropBank
SQuAD / TriviaQA / NQ	Answer spans / evidence	Reading comprehension QA	Public	Good QA baselines but not semantically structured li

Table 3. Dataset comparison for SRL-QA-oriented research.

Open Challenges and Research Gaps

The first major gap is generalization beyond local, contiguous event slots. Discontinuous arguments, nested roles, cross-sentence dependencies, and discourse-conditioned answers remain awkward for most PropBank-based QA pipelines. The current project's preprocessing encountered the same issue, confirming that classical extractive assumptions still exclude a meaningful part of the underlying semantic space.

A second gap concerns calibration and abstention. Most SRL-QA systems produce a best span even when the question is poorly grounded, yet practical deployments need confidence thresholds and explicit refusal behavior. Recent LLM work has made this problem harder to ignore because fluent answers can obscure structural uncertainty.

Third, there is still no standard benchmark for multi-predicate, multi-sentence SRL-QA. Existing resources either remain predicate-local or migrate to broader QA-style semantic representations without preserving consistent SRL-QA evaluation. This makes cross-paper comparison difficult and motivates new benchmark design.

Finally, multilingual and low-resource SRL-QA remains open. PropBank-style annotation has spread across languages, but QA-based semantic interfaces are still largely Anglocentric. Future work will depend on better cross-lingual transfer, domain adaptation, and question-generation strategies tied to language-specific role realizations.

A subtler gap concerns evaluation philosophy. Exact match and token overlap are useful, but they do not reveal whether the system chose the right predicate frame, whether the answer is semantically justified, or whether the returned span would remain valid under paraphrase. Future SRL-QA benchmarks will need faithfulness-oriented metrics that are sensitive to semantic explanation, not just lexical similarity.

There is also a tooling gap. Academic papers increasingly assume access to large pretrained models, but reproducible semantic pipelines still depend on careful corpus interfaces, robust pointer resolution, and interoperable annotation formats. NLTK's PropBank wrapper remains valuable precisely because it makes these low-level engineering questions visible.

Conclusion

The literature shows that SRL-based QA is best understood as a family of structured reasoning strategies rather than a narrow benchmark niche. Whether the goal is annotation, parsing, alignment, extraction, or explanation, the shared insight is that questions and semantic roles can serve as mutually interpretable views of event meaning.

PropBank remains a particularly effective substrate for classical SRL-QA because it balances coarse label consistency with predicate-specific interpretability. The next research frontier lies in bringing that explicit structure into closer contact with long-context reasoning, calibrated uncertainty, and LLM-era semantic grounding rather than discarding structure altogether.

For teaching, reproducibility, and transparent experimentation, this combination remains unusually valuable. A student or researcher can inspect the frame files, reconstruct the argument spans, trace the generated question, and examine the exact failure case. Few contemporary QA paradigms retain that degree of inspectability while still supporting a meaningful end task.

Extended Discussion

One underappreciated strength of SRL-QA is pedagogical transparency. Because the pipeline decomposes into predicate detection, role recovery, question mapping, and answer extraction, it gives researchers a clean way to discuss where semantic understanding actually enters the system. This is harder to do with end-to-end answer generators whose intermediate decisions are largely latent.

Another strength is compatibility with downstream auditing. In legal, financial, medical, and public-sector settings, it is often not enough to retrieve a phrase; the analyst also needs to know whether that phrase was interpreted as agent, patient, location, or cause. SRL-QA surfaces that information natively.

The survey also suggests a strategic research stance. Instead of asking whether SRL should compete directly with the largest generative QA systems, it may be more productive to ask where explicit semantics offers unique leverage. The answer is usually in explanation, controlled querying, and event-structured analysis rather than open-ended encyclopedic QA.

Finally, the literature points toward a synthesis rather than a replacement story. Question-based semantics has made semantic annotation more accessible; neural modeling has made role recovery more robust; and LLMs are now creating new opportunities for frame-aware reasoning and adjudication. The field is moving toward hybrid systems that preserve structure while gaining flexibility.

That synthesis is the main intellectual takeaway of the survey. Explicit semantic roles continue to matter because they provide a stable unit of reasoning that can be inspected, aligned, composed, and constrained. In a research landscape increasingly dominated by large, opaque models, that stability is a meaningful asset.

Final Reflection

A final lesson from the literature is methodological humility: semantic resources such as PropBank, QA-SRL, and FrameNet do not eliminate the need for better models, but they do provide stable interfaces for asking what a model actually knows. That is a rare and increasingly valuable property in modern NLP.

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